NuNo-KD: Spectral-Aware Knowledge Distillation via Nuclear Norm Matching

Anonymous ACL submission

Abstract

Knowledge distillation (KD) compresses large language models by transferring teacher behavior to a smaller student, yet most existing on-policy methods supervise the student exclusively through output-layer probabilities, leaving intermediate hidden-state geometry unconstrained. Under capacity mismatch, this creates a structural blind spot: the student can match the teacher's output distribution while collapsing its hidden-state diversity into a few dominant directions, losing intermediate structure critical for multi-step reasoning. We propose NuNo-KD, which augments a standard on-policy KD backbone with Nuclear Norm Matching (NNM), a lightweight representation-level objective that encourages the student to match the teacher's layer-level spectral magnitude (the nuclear norm) at mid-layers, without performing restrictive pointwise feature regression. To stabilize mini-batch spectral estimates, we anchor each batch against teacher-derived centroids that summarize the global representation distribution. Experiments on diverse reasoning benchmarks show consistent average gains over strong on-policy and off-policy baselines, with layer-wise spectral analysis confirming that gains reflect genuine improvements in intermediate representation quality rather than surface-level output calibration.

1 Introduction

While large language models (LLMs) excel at complex reasoning, their prohibitive size necessitates Knowledge Distillation (KD) (Hinton et al., 2015). However, most existing on-policy methods (Agarwal et al., 2024; Ko et al., 2024; Shin et al., 2026) share a critical blind spot: they supervise exclusively through output-layer probabilities, leaving intermediate representations unconstrained. This blindness is not incidental but structural: regardless of whether supervision uses forward KL, reverse KL, or skewed variants, the student's hidden states enter the objective only through the final-layer logits that produce the next-token distribution. Consequently, a student whose intermediate representations collapse into a low-dimensional subspace incurs no penalty, as long as the surviving directions span the output projection—yet Arefin et al. (2025) and Skean et al. (2025) establish that mid-layer matrix entropy is functionally linked to reasoning: collapsing it impairs multi-step inference, and reasoning-focused fine-tuning reliably raises it. A student model can therefore perfectly mimic the teacher's output distribution while entirely losing the internal geometric diversity that supports complex reasoning.

Recent KD methods targeting token-level entropy (Kim and Baek, 2026; Jin et al., 2026) cannot resolve this, as token-level Shannon entropy and the matrix entropy of intermediate states are not interchangeable. For instance, explicitly targeting high-entropy tokens in TSD-KD (Kim and Baek, 2026) fails to mitigate mid-layer representation collapse, evidenced by near-zero matrix entropy (Figure 1). Conversely, naively regularizing the student toward maximum matrix entropy is also flawed, as mid-layer compression is a functional feature of well-trained models (Queipo-de Llano et al., 2026; Skean et al., 2025). Thus, we argue that under capacity mismatch, the functionally correct objective is to match a teacher-defined spectral target, not artificially maximize it.

We propose **Nu**clear **No**rm Matching **K**nowledge **D**istillation (NuNo-KD), augmenting an on-policy objective (Ko et al., 2024) with a penalty matching the nuclear norms—the tightest convex surrogate for rank (Recht et al., 2010)—of intermediate hidden states. NuNo-KD relies on two principles: targeting mid-layers where output supervision is weakest and stabilizing estimates with teacher-derived centroid anchors. Crucially, to ensure computational tractability, NuNo-KD bypasses explicit Singular Value Decomposition

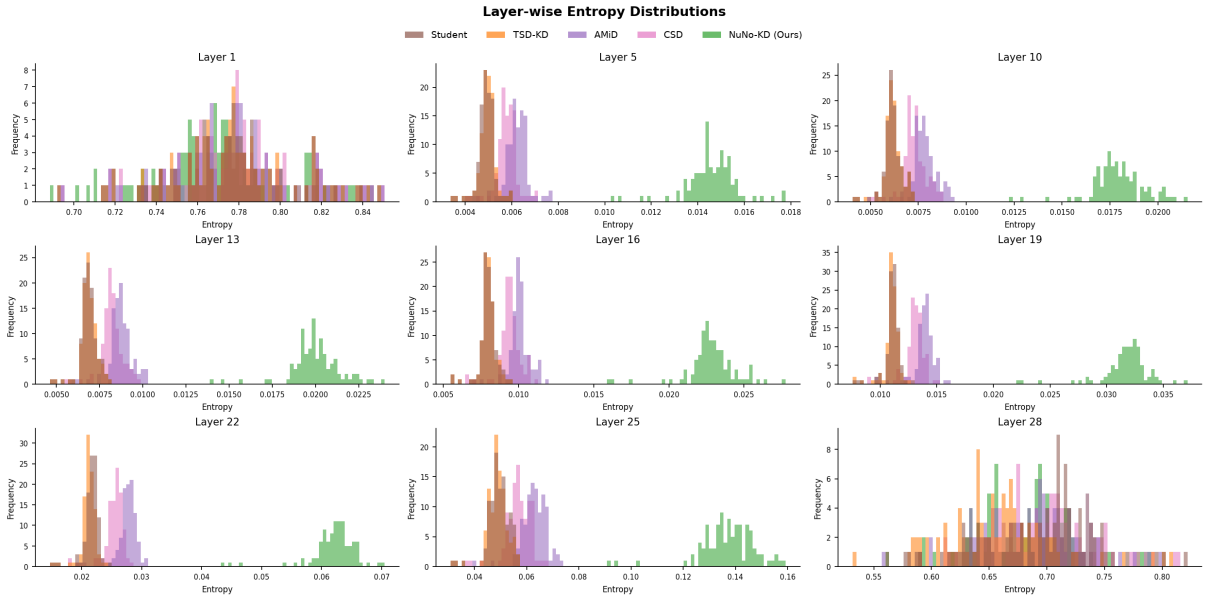


Figure 1: Layer-wise matrix entropy distributions of the Qwen2.5-1.5B-Instruct student, computed exclusively on raw hidden states (H_s) to verify intrinsic improvements. In the mid-layer compression valley (layers 2–26; layers 5–25 shown; NNM supervises layers 5, 11, 17, 23), output-only baselines overlap or shift only marginally from Student. NuNo-KD, however, exhibits a consistent rightward shift (broader spectral spread). Full distributions in Appendix C.

(SVD) entirely via random projections and a hardware-friendly Newton–Schulz iteration. Experiments across two model families show that NuNo-KD achieves consistent average gains over strong baselines on diverse reasoning benchmarks.

Contributions.

- We identify that output-only KD structurally ignores intermediate geometry, as token-level and matrix-level entropies are not interchangeable.
- We introduce NuNo-KD, a centroid-anchored, mid-layer-focused objective that matches the teacher's layer-level spectral magnitude via scalable, hardware-efficient nuclear norm matching.
- We demonstrate consistent average gains over state-of-the-art baselines across diverse reasoning benchmarks, without introducing inference overhead.

2 Related Work

Output-level distillation. Off-policy KD trains the student on fixed data (Hinton et al., 2015; Kim and Rush, 2016; Magister et al., 2023) and suffers from a training-inference mismatch. On-policy KD instead trains on student samples, using reverse KL (MiniLLM; Gu et al., 2024), generalized divergences (GKD; Agarwal et al., 2024), skew KL (DistiLLM and DistiLLM-2; Ko et al., 2024, 2025), refined objectives (CSD and AMiD; Kim et al., 2026; Shin et al., 2026), or high-entropy token selection (TSD-KD; Kim and Baek, 2026). All of them supervise only output probabilities; we adopt the skew KL objective of Ko et al. (2024) as our backbone.

Representation-level distillation. Prior work regresses projected hidden states (Romero et al., 2015; Sun et al., 2019, 2020), matches pairwise relations (Park et al., 2019; Wang et al., 2020), aligns CKA across hidden widths (Dasgupta and Cohn, 2025), or projects hidden states between models (Zhang et al., 2024). These methods transfer the teacher's features or their pairwise structure. Under a large capacity gap, such pointwise or pairwise targets can over-constrain the student, forcing it to reproduce fine-grained teacher geometry that it lacks the capacity to represent.

Representational collapse. Representational collapse, the contraction of embeddings into a low-dimensional subspace, is well studied in self-supervised vision (Jing et al., 2022; Grill et al., 2020; Zbontar et al., 2021; Bardes et al., 2022), but its remedies are teacher-free and do not transfer directly to distillation. In language models, Barbero et al. (2024) document last-layer collapse, and Queipo-de Llano et al. (2026), Skean et al. (2025), and Arefin et al. (2025) link mid-layer collapse to

reasoning, which motivates our mid-layer targets (Section 3).

3 Method: Nuclear Norm Matching Knowledge Distillation

Setup and Notation. Let the teacher have L_t layers and hidden size d_t, and the student L_s layers and hidden size d_s, with $d_s \leq d_t$. We select n matched layer pairs

$$\mathcal{P} = \{(L_s^{(i)}, L_t^{(i)})\}_{i=1}^{n}, \tag{1}$$

where $i \in \{1, \ldots, n\}$ indexes the pairs, $L_s^{(i)} \in \{1, \ldots, L_s\}$ is the student layer index, and $L_t^{(i)} \in \{1, \ldots, L_t\}$ is the corresponding teacher layer index. Pairs are chosen so that relative depths are aligned: $L_s^{(i)}/L_s \approx L_t^{(i)}/L_t$. NNM operates in an on-policy regime: for each prompt $x \sim \mathcal{D}$ the student generates a response $y_s \sim p_s(\cdot \mid x)$, and both teacher and student are evaluated on the same sequence (x, y_s).

After removing prompt and padding positions, let T denote the number of response tokens in the current mini-batch. We collect response-token hidden states at each matched layer:

$$H_s^{(i)} \in \mathbb{R}^{T \times d_s}, \qquad H_t^{(i)} \in \mathbb{R}^{T \times d_t}, \tag{2}$$

where the superscript $i \in \{1, \ldots, n\}$ refers to the i-th matched pair in $\mathcal{P}$.

Because $d_s \leq d_t$, comparing $H_s^{(i)}$ and $H_t^{(i)}$ directly is ill-defined. We introduce a learnable linear projector $W_{s \to t} \in \mathbb{R}^{d_s \times d_t}$ (one per matched pair) and define the projected student matrix

$$\bar{H}_s^{(i)} = H_s^{(i)} W_{s \to t} \in \mathbb{R}^{T \times d_t}. \tag{3}$$

Each $W_{s \to t}$ receives gradients only from $\mathcal{L}_{\text{NNM}}$ and is discarded at inference.

3.1 Motivation: The Output-Only Blind Spot

(A) Limitations of output KD. Most output-level KD objectives (forward KL, reverse KL, skew KL, ...) reduce to the essential form

$$\mathcal{L}_{\text{KD}} = \mathbb{E}_{(x,y)\sim\mathcal{D}} \sum_{\tau} \mathcal{D}[p_t(\cdot \mid x, y_{<\tau}),\ p_s(\cdot \mid x, y_{<\tau})] \tag{4}$$

where τ indexes token positions, $y_{<\tau}$ is the prefix up to position τ, and $\mathcal{D}[\cdot, \cdot]$ is a divergence between teacher and student next-token distributions. Whether y is drawn from the teacher's distribution (off-policy) or the student's (on-policy) does not affect the argument: in either case, the student's hidden states enter Eq. (4) only through the final-layer logits that produce p_s. A student whose intermediate hidden states $H_s^{(i)}$ are confined to a low-dimensional subspace incurs no penalty under $\mathcal{L}_{\text{KD}}$ as long as the surviving directions span the output projection. Output-only KD is structurally blind to mid-layer spectral geometry.

Figure 1 provides empirical support for this phenomenon. TSD-KD, CSD, and AMiD—despite achieving the strongest output-level accuracy among prior baselines (Table 1)—produce mid-layer matrix entropy distributions that exhibit only a marginal, near-zero shift from Student. The gap between their downstream accuracy and their spectral profile is precisely the dissociation that Eq. (4) predicts: output objectives can improve p_s while barely affecting $\mathcal{H}_{\text{mat}}(H_s^{(i)})$.

(B) Token-level entropy and matrix entropy are not the same. Empirical evidence directly links mid-layer matrix entropy to reasoning capabilities. Arefin et al. (2025) show that it controls multi-step reasoning success in GPT-2, where preventing collapse directly rescues performance. Furthermore, Skean et al. (2025) find that fine-tuning models for mathematical reasoning substantially raises mid-layer matrix entropy. Despite this, recent reasoning-oriented distillation methods (Kim and Baek, 2026; Jin et al., 2026) emphasize token-level Shannon entropy over output logits. We argue that conflating these two distinct quantities leaves a critical blind spot. We distinguish:

Token-level Shannon entropy (Shannon, 1948). $\mathcal{H}_{\text{tok}}(\tau) = -\sum_{v \in \mathcal{V}} p_s(v \mid x, y_{<\tau}) \log p_s(v \mid x, y_{<\tau})$, with vocabulary $\mathcal{V}$. This is a scalar property of one position's output distribution, computed from logits.

Matrix entropy (Arefin et al., 2025). Let $H \in \mathbb{R}^{T \times d}$ (e.g., $H_s^{(i)}$) have singular values $\sigma_1 \geq \cdots \geq \sigma_r$, $r = \min(T, d)$, and $\tilde{\sigma}_k^2 = \sigma_k^2 / \sum_j \sigma_j^2$. The matrix entropy is

$$\mathcal{H}_{\text{mat}}(H) = -\sum_{k=1}^{r} \tilde{\sigma}_k^2 \log \tilde{\sigma}_k^2, \tag{5}$$

equivalently the Shannon entropy (Shannon, 1948) over the spectrum of the normalized Gram matrix $\tilde{G} = H^\top H / \|H\|_F^2$.

Figure 1 strongly supports this decoupling. For instance, **TSD-KD** (Kim and Baek, 2026) attempts

to improve reasoning by selectively distilling high-entropy tokens. However, its mid-layer matrix entropy distributions overlap with the collapsed Student. Concentrating supervision exclusively on output-level token diversity shows limited ability to translate into structural geometric diversity at mid-layers. If the intermediate representations collapse, the student inevitably loses the geometric capacity to genuinely replicate complex reasoning trajectories.

(C) Proposed solution. We argue that in addition to output-level KD, distillation must explicitly match a teacher-derived spectral statistic at mid-layers. Rather than maximizing or ignoring this profile, we require a statistic that is numerically stable, convex in the hidden-state matrix, and computationally tractable. Section 3.2 demonstrates that the nuclear norm satisfies all three criteria.

3.2 Spectral Target: Nuclear Norm Matching

To optimize the student's spectral spread efficiently, we require a target metric that reliably captures intermediate geometry without disrupting the training dynamics.

Why nuclear norm rather than matrix entropy directly? For a given representation matrix $F \in \mathbb{R}^{A\times B}$, the nuclear norm is defined formally as the sum of its singular values:

$$\|F\|_* = \sum_{k=1}^{\min(A,B)} \sigma_k(F), \tag{6}$$

where $\sigma_1(F) \geq \sigma_2(F) \geq \cdots \geq 0$ denote the singular values of F.

The nuclear norm is the tightest convex surrogate for matrix rank (Recht et al., 2010). Eq. (6) captures essential spectral properties without inheriting matrix entropy's pathologies. Specifically, matrix entropy is non-convex, and its derivative in each $\tilde{\sigma}_k^2$, $-(\log \tilde{\sigma}_k^2 + 1)$, diverges as $\tilde{\sigma}_k \to 0$ during representation collapse. Computationally, exact nuclear norm calculation requires a singular value decomposition (SVD), which matrix entropy also needs and which parallelizes poorly on GPUs. To make NuNo-KD practically scalable, we completely bypass explicit SVD. Instead, we efficiently approximate the nuclear norm via a hardware-friendly Newton–Schulz iteration (Higham, 1986; Kim and Oh, 2026) that relies entirely on fast matrix multiplications (see Appendix A.1 for implementation details). Thus, it provides a stable and computationally feasible distillation target.

3.3 Layer Selection Range

We space the n matched layers evenly in the layer-selection range $[0.20, 0.85]$ of relative depth (layers 5, 11, 17, 23 of our 28-layer student; depths 0.18–0.82). These bounds are motivated by the distinct functional properties of the corresponding layer strata.

The **lower bound** of 0.20 excludes the detokenization stage identified by Skean et al. (2025). Early layers primarily convert raw subword tokens into a basic embedding space and are highly sensitive to noise. Random token substitutions can inflate early-layer entropy significantly, whereas deeper layers remain robust. Imposing spectral supervision in this early stratum therefore risks amplifying noise-driven entropy fluctuations rather than providing a meaningful structural signal.

The **upper bound** of 0.85 excludes the final layers for a structural reason: this region is already explicitly supervised by the output-level objective ($\mathcal{L}_{\text{SKL}}$); adding $\mathcal{L}_{\text{NNM}}$ here provides redundant signal and risks distorting the directions the student must align with its own output projection.

Together, these bounds keep the supervised layers inside the compression valley (Queipo-de Llano et al., 2026; Skean et al., 2025)—a theoretically grounded representational stratum where semantic and multi-step reasoning computations are localized (Arefin et al., 2025).

Centroid-Anchored Spectral Reference. A naive implementation would match $\|\bar{H}_s^{(i)}\|_*$ and $\|H_t^{(i)}\|_*$ directly within each mini-batch. We avoid this for two reasons. First, the nuclear norm of a mini-batch matrix is a local snapshot: a batch may appear spectrally diverse while the underlying representation distribution is globally compressed, and vice versa. Second, mini-batch estimates are noisy at the modest batch sizes typical of LLM fine-tuning, producing a high-variance training signal.

We address both by anchoring each batch against a stable teacher-derived spectral reference. For each matched teacher layer $L_t^{(i)}$, we run a teacher pre-pass over a subset of the training data and estimate K centroids via an online K-means-style update (Sculley, 2010):

$$C_t^{(i)} = [\mu_{i,1}; \ldots; \mu_{i,K}] \in \mathbb{R}^{K\times d_t}, \tag{7}$$

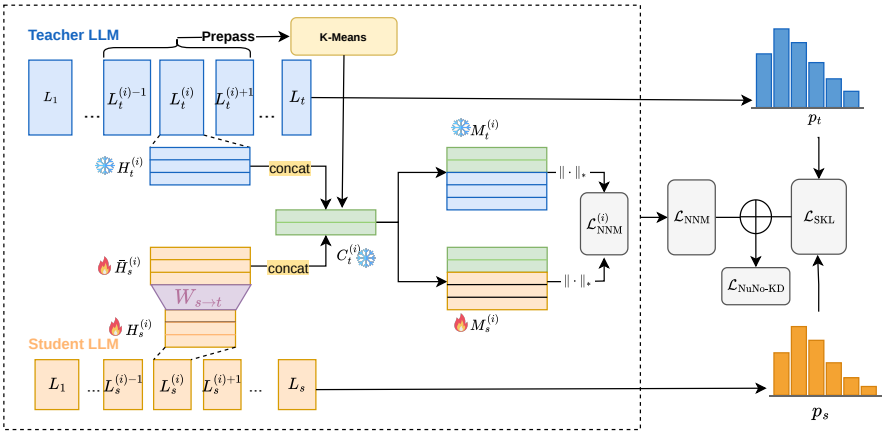


Figure 2: Overview of NuNo-KD. Precomputed teacher centroids $C_t^{(i)}$ are concatenated with teacher and projected student hidden states to form anchored matrices $M_t^{(i)}$ and $M_s^{(i)}$. The layer-wise losses $\mathcal{L}_{\text{NNM}}^{(i)}$ (the i-th summand of Eq. (10)) match their log nuclear norms and are averaged into $\mathcal{L}_{\text{NNM}}$, which supplements the output-level SKL objective ($\mathcal{L}_{\text{NuNo-KD}} = \mathcal{L}_{\text{SKL}} + \lambda \mathcal{L}_{\text{NNM}}$). The random projection R (Eq. (9)) is omitted for clarity.

where $[\cdot\,;\cdot]$ stacks rows ($\mu_{i,k} \in \mathbb{R}^{1\times d_t}$). The centroids are computed once and fixed throughout student training, serving as a teacher-side spectral memory—a coarse summary of the directions the teacher uses across the training distribution. At each training step we concatenate the same centroid matrix with both batches:

$$M_s^{(i)} = \left[C_t^{(i)}; \bar{H}_s^{(i)}\right], \quad M_t^{(i)} = \left[C_t^{(i)}; H_t^{(i)}\right] \tag{8}$$

each with $K+T$ rows. Because both sides share the same anchors, the nuclear norms of $\bar{M}_s^{(i)}$ and $\bar{M}_t^{(i)}$ in Eq. (10) measure how the current on-policy batch expands a common teacher-defined spectral scaffold. This makes the comparison teacher-referenced and substantially reduces mini-batch variance. Furthermore, to alleviate the computational bottleneck associated with spectral operations on high-dimensional hidden states, we introduce a fixed random projection matrix $R \in \mathbb{R}^{d_t \times d'}$ ($d' \ll d_t$). We right-multiply both joint matrices by R to obtain their dimensionally reduced counterparts:

$$\bar{M}_s^{(i)} = M_s^{(i)} R, \quad \bar{M}_t^{(i)} = M_t^{(i)} R. \tag{9}$$

By the Johnson–Lindenstrauss lemma (Johnson and Lindenstrauss, 1984), $d' = O(\log(K+T)/\epsilon^2)$ preserves all pairwise row distances up to a factor of $1\pm\epsilon$, cutting the cost of the spectral computation.

Nuclear Norm Matching Loss. Concretely, the total NNM loss averages the squared difference between the log-transformed nuclear norms across all n matched layers:

$$\mathcal{L}_{\text{NNM}} = \frac{1}{n}\sum_{i=1}^{n}\left(\ln\left(\|\bar{M}_s^{(i)}\|_*\right) - \ln\left(\|\bar{M}_t^{(i)}\|_*\right)\right)^2 \tag{10}$$

This squared-distance formulation penalizes spectral spread that is strictly too low or too high across the compression valley.

3.4 Full Training Objective

As shown in Figure 2, NNM supplements a token-level KD backbone—the skewed forward KL (SKL) divergence introduced in DistiLLM (Ko et al., 2024). By evaluating the loss on student-generated trajectories, this approach fully exploits the on-policy distillation regime. SKL stabilizes pure forward KL by smoothing the comparison distribution with a small contribution from the teacher:

$$D_{\text{SKL}}^{(\alpha)}(p_t \,\|\, p_s) = \text{KL}\big(p_t \,\big\|\, (1-\alpha)\, p_s + \alpha\, p_t\big) \tag{11}$$

with skewness coefficient $\alpha \in (0,1)$. The on-policy KD loss sums it over the positions of y_s:

$$\mathcal{L}_{\text{SKL}} = \mathbb{E}_{x\sim\mathcal{D},\, y_s\sim p_s(\cdot|x)} \sum_{\tau} D_{\text{SKL}}^{(\alpha)}\big(p_t^\tau \,\|\, p_s^\tau\big), \tag{12}$$

where p_t^τ, p_s^τ condition on $(x, y_{s,<\tau})$; we set $\alpha = 0.1$. The complete training objective is:

$$\mathcal{L}_{\text{NuNo-KD}} = \mathcal{L}_{\text{SKL}} + \lambda\, \mathcal{L}_{\text{NNM}}. \tag{13}$$

$\mathcal{L}_{\text{SKL}}$ aligns the student's output distribution with the teacher's on student-generated trajectories; $\mathcal{L}_{\text{NNM}}$ aligns the log nuclear norm of the student's mid-layer representations with the teacher's on the same trajectories.

Method	GSM8K	GSM-Plus	MATH	MBPP	SciQ	MMLU-STEM	MMLU-Pro-Math	BBH	Avg.
Qwen2.5-14B-Instruct (Teacher)	80.3	59.7	21.7	78.9	86.4	70.5	77.6	60.8	67.0
Qwen2.5-1.5B-Instruct (Student)	57.1	38.8	16.9	38.4	85.5	49.5	34.4	36.5	44.6
Sequence-Level KD	56.5	38.2	16.5	40.6	87.9	48.9	32.6	36.6	44.7
Supervised KD	56.3	37.8	17.9	41.0	86.6	48.4	35.3	36.4	45.0
DistiLLM	57.2	38.7	18.6	42.1	85.7	47.8	35.2	36.5	45.2
Speculative KD	58.0	39.2	18.5	41.6	86.7	47.8	34.8	37.2	45.5
MiniLLM	57.7	39.7	17.8	42.2	84.1	48.2	36.2	36.0	45.2
GKD ($\beta = 0.9$)	57.9	39.9	18.1	41.8	86.6	47.7	35.7	36.2	45.5
TSD-KD	60.1	40.5	26.1	42.1	**93.0**	50.0	36.9	40.2	48.6
CSD	61.5	41.5	29.3	43.8	81.9	51.3	**44.4**	42.1	49.5
AMiD	61.1	42.1	28.3	43.6	86.7	51.6	40.0	42.0	49.4
NuNo-KD (Ours)	**65.2**	**46.8**	**32.2**	**44.4**	86.2	**51.7**	42.0	**44.9**	**51.7**

Table 1: Performance comparison on Qwen2.5-14B-Instruct (Teacher) → Qwen2.5-1.5B-Instruct (Student). **Bold**: best, underlined: second best.

Method	GSM8K	GSM-Plus	MATH	MBPP	MMLU-STEM	Avg.
Gemma-2-9B-IT (Teacher)	57.7	40.8	25.8	40.0	50.2	42.9
Gemma-2-2B-IT (Student)	53.7	36.2	16.1	39.2	25.5	34.1
Sequence-Level KD	53.4	33.8	17.7	39.2	25.9	34.0
Supervised KD	53.6	36.8	17.9	39.0	25.5	34.6
DistiLLM	54.3	35.8	17.8	39.6	25.2	34.5
MiniLLM	54.4	36.8	17.5	**40.6**	25.9	35.0
GKD ($\beta = 0.9$)	54.1	36.9	17.2	40.0	26.2	34.9
TSD-KD	55.3	37.9	18.9	40.4	27.0	35.9
CSD	47.2	32.0	19.6	31.6	47.4	35.6
AMiD	49.6	33.8	20.5	33.4	44.7	36.4
NuNo-KD (Ours)	**56.0**	**38.7**	**23.6**	39.4	**49.3**	**41.4**

Table 2: Performance comparison on Gemma-2-9B-IT (Teacher) → Gemma-2-2B-IT (Student). **Bold**: best, underlined: second best.

4 Experiments

4.1 Experimental Settings

Models. Our primary experiments distill Qwen2.5-14B-Instruct into Qwen2.5-1.5B-Instruct (Yang et al., 2024). To demonstrate generalizability, we also distill Gemma-2-9B-IT into Gemma-2-2B-IT (Gemma Team et al., 2024).

Baselines. We compare against seven on-policy methods, which train the student on its own outputs—DistiLLM (Ko et al., 2024), MiniLLM (Gu et al., 2024), GKD (Agarwal et al., 2024) ($\beta = 0.9$), Speculative KD (Xu et al., 2025), TSD-KD (Kim and Baek, 2026), AMiD (Shin et al., 2026), and CSD (Kim et al., 2026)—and two off-policy methods trained on teacher-generated responses, Supervised KD (Hinton et al., 2015) and Sequence-Level KD (Kim and Rush, 2016).

Training. All methods train on the approximately 78K UltraInteract (Yuan et al., 2025) instances released by TSD-KD (Kim and Baek, 2026), with student-generated responses for on-policy methods and teacher-generated responses for off-policy ones; hyperparameters are in Appendix A.2.

Evaluation. We evaluate all models across a broad suite of reasoning benchmarks spanning four categories: mathematical reasoning, covering GSM8K (Cobbe et al., 2021), GSM-Plus (Li et al., 2024), MATH (Hendrycks et al., 2021b), and MMLU-Pro-Math (Wang et al., 2024); STEM and scientific reasoning, comprising MMLU-STEM (Hendrycks et al., 2021a) and SciQ (Welbl et al., 2017); program synthesis, evaluated on MBPP (Austin et al., 2021); broad reasoning, assessed via BBH (Suzgun et al., 2023) with chain-of-thought prompting. Accuracy is reported for all benchmarks following their respective standard evaluation protocols. Moreover, following Skean et al. (2025), we evaluate the learned representations layer by layer with four metrics: the size-normalized nuclear norm $\|H\|_*/\sqrt{Nd}$ of N token states of width d, effective rank (Roy and Vetterli, 2007), matrix entropy (Sanchez Giraldo et al., 2015), and curvature (Hosseini and Fedorenko, 2023).

4.2 Main Results

NuNo-KD achieves the best overall performance, leading on six of eight benchmarks. On Qwen2.5 (Table 1), NuNo-KD ranks first on six of eight benchmarks and achieves the highest average accuracy (51.7), 2.2 points above the strongest prior baseline, CSD (49.5); it gains 7.1 points over Student, whereas CSD gains 4.9. Its largest margins over the best baseline are on GSM8K (+3.7), GSM-Plus (+4.7), MATH (+2.9), and BBH (+2.8), and it gains 7.6 over Student on MMLU-Pro-Math.

The gains generalize across model families, with an even wider margin on Gemma-2. On Gemma-2 (Table 2), NuNo-KD reaches 41.4 average accuracy, 5.0 points above AMiD and 7.3

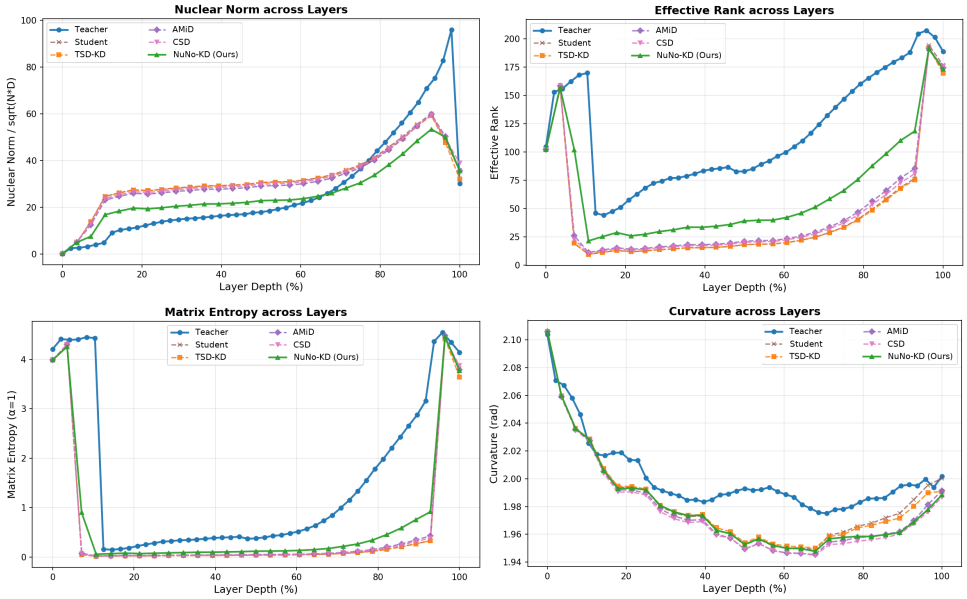


Figure 3: Layer-wise spectral and geometric metrics, computed exclusively on raw hidden states (H_s) to verify intrinsic improvements. While output-only baselines overlap or shift only marginally from Student, NuNo-KD explicitly elevates effective rank and matrix entropy across the targeted $[0.20, 0.85]$ depth range toward the teacher.

Method	GSM8K	GSM-Plus	MATH	MBPP	SciQ	MMLU-STEM	Avg.
Qwen2.5-1.5B-Instruct (Student)	57.1	38.8	16.9	38.4	85.5	49.5	47.7
w/o centroid anchoring	57.5	39.8	0.7	4.2	**88.1**	**52.8**	40.5
Frozen random $W_{s\to t}$	52.8	34.3	29.3	43.8	83.7	34.2	46.4
Frobenius regression	50.3	32.5	29.0	34.6	81.5	44.0	45.3
Gram	**72.7**	**52.1**	29.1	**45.8**	68.4	50.6	53.1
CKA	68.5	49.6	25.7	42.0	69.2	50.3	50.9
Direct spectrum	62.8	45.5	25.2	14.4	63.1	51.3	43.7
NuNo-KD (Ours)	65.2	46.8	**32.2**	44.4	86.2	51.7	**54.4**

Table 3: Component ablations (second block) and alternative hidden-state objectives (third block) on Qwen2.5-14B-Instruct → Qwen2.5-1.5B-Instruct. Avg. is over the six benchmarks. Setups in Appendix A.3. **Bold**: best among the distilled students.

above Student, and ranks first on four of five benchmarks. It raises MMLU-STEM from 25.5 to 49.3 and MATH from 16.1 to 23.6 (3.1 above AMiD, the best baseline), while also improving both GSM benchmarks and keeping MBPP at Student level (39.4 vs. 39.2), whereas CSD and AMiD gain on MMLU-STEM but fall below Student on both GSM benchmarks and MBPP.

4.3 Ablation Study

Unless otherwise noted, all ablations use the Qwen2.5-14B-Instruct → Qwen2.5-1.5B-Instruct teacher–student pair and are evaluated on six benchmarks: GSM8K, GSM-Plus, MATH, MBPP, SciQ, and MMLU-STEM. Any deviations from this setup are explicitly noted for the corresponding ablation.

Layer-wise representation analysis. All metrics are computed on raw student hidden states H_s, before $W_{s\to t}$, so they reflect the student rather than the projector. As shown in Figures 1 and 3, CSD, AMiD, and TSD-KD barely change mid-layer entropy and effective rank, whereas NuNo-KD raises both, including at unsupervised layers of the compression valley (Appendix C). The size-normalized nuclear norm in Figure 3 still grows with activation scale: the baselines lie above the teacher over roughly the first 65% of depth, and NuNo-KD lowers it toward the teacher. Dividing by $\|H\|_F$ reverses the order (Appendix B): NuNo-KD shrinks the activation scale while spreading the spectrum, so these spectral changes cannot be explained by uniform rescaling alone. Curvature barely differs across students, which all stay below the teacher.

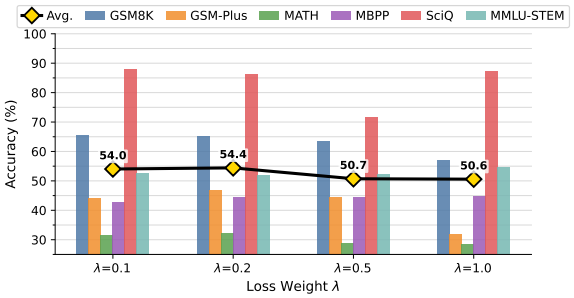


Figure 4: Sensitivity to the loss weight λ.

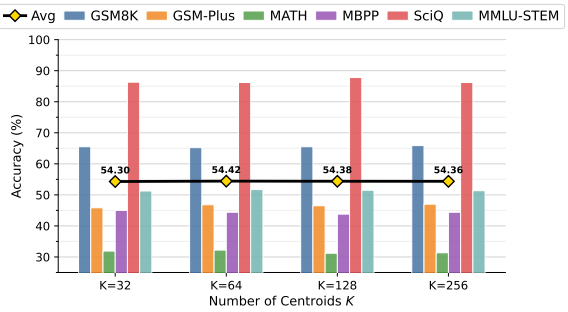


Figure 5: Robustness to number of centroids K.

Range	GSM8K	GSM-Plus	MATH	MBPP	SciQ	MMLU-STEM	Avg.
$[0.10, 0.90]$	**66.0**	46.7	31.1	**44.6**	85.9	51.3	54.3
$[0.20, 0.85]$	65.2	**46.8**	**32.2**	44.4	86.2	**51.7**	**54.4**
$[0.70, 1.00]$	64.1	46.3	30.9	44.1	**87.0**	51.2	53.9

Table 4: Effect of the layer selection range. **Bold**: best.

n	GSM8K	GSM-Plus	MATH	MBPP	SciQ	MMLU-STEM	Avg.
2	64.3	46.3	32.0	43.8	86.4	51.4	54.0
4	**65.2**	**46.8**	**32.2**	**44.4**	86.2	**51.7**	**54.4**
8	**65.2**	46.5	31.3	**44.4**	**86.5**	51.2	54.2

Table 5: Effect of the layer count n. **Bold**: best.

Component ablations and alternative objectives. Table 3 shows that every component matters and that no alternative objective is uniformly better. Removing centroid anchoring is the most damaging: the average falls below Student, with MATH and MBPP collapsing while SciQ and MMLU-STEM stay high, suggesting that a batch-local target is unstable for long-form generation. A frozen random $W_{s \to t}$ and Frobenius regression also fall below Student on average, raising MATH but hurting knowledge tasks: the projector must be learned, and point-wise matching fits poorly under a large capacity gap. Gram is stronger on GSM8K, GSM-Plus, and MBPP, but Gram and CKA lose about 17 points on SciQ relative to Student, and direct spectrum matching degrades MBPP and SciQ; constraining the full relational structure or every singular value likely leaves the student too little room to adapt. NuNo-KD constrains a single layer-level statistic and is the only objective that improves over Student on all six benchmarks, so its 1.3-point lead over Gram in average reflects a balanced profile rather than gains on a few tasks.

Sensitivity to the loss weight and the number of centroids. Figure 4 reports average accuracy over the six benchmarks as $\lambda \in \{0.1, 0.2, 0.5, 1.0\}$. Performance peaks at $\lambda = 0.2$ and drops sharply at $\lambda \geq 0.5$, so the spectral term works best as a regularizer that supplements, rather than dominates, the output-level backbone; even $\lambda = 0.1$ (54.0) beats the best baseline (AMiD, 52.2). In contrast, Figure 5 shows that NuNo-KD is insensitive to the number of centroids: accuracy stays within 54.30–54.42 for $K \in \{32, 64, 128, 256\}$, so a coarse summary of the teacher's representation distribution suffices to anchor the comparison.

Sensitivity to the layer selection range. Table 4 varies the layer-selection range with $n = 4$. The motivated range $[0.20, 0.85]$ scores highest (54.4) and widening it to $[0.10, 0.90]$ costs only 0.1 points, whereas restricting supervision to the late layers $[0.70, 1.00]$ gives the lowest score (53.9), likely because these layers are already supervised by $\mathcal{L}_{\text{SKL}}$. All three ranges still exceed the best baseline.

Sensitivity to the number of supervised layers. Table 5 evaluates $n \in \{2, 4, 8\}$ supervised layers evenly spaced in $[0.20, 0.85]$. Average accuracy peaks at $n = 4$. With $n = 2$, the largest drops are on GSM8K (-0.9) and MBPP (-0.6), suggesting that two anchors leave parts of the compression valley under-constrained. With $n = 8$, GSM8K and MBPP match $n = 4$ but MATH drops by 0.9; since the effect of NNM already propagates to unsupervised layers (Appendix C), densely spaced layers add overlapping constraints without improving coverage. Even $n = 2$ beats every baseline.

5 Conclusion

We introduced NuNo-KD, which augments on-policy distillation with Nuclear Norm Matching to match the teacher's mid-layer spectral magnitude—a structural blind spot of output-only objectives. Experiments across reasoning benchmarks show consistent average gains, and layer-wise analyses indicate richer intermediate representations rather than better output calibration alone.

Limitations

First, due to computational budget constraints, our experiments are limited to moderately sized teachers (14B and 9B parameters), and we were unable to evaluate NuNo-KD with larger teacher models; whether the method's benefits extend to such settings remains an open empirical question. Second, NuNo-KD assumes access to a single fixed teacher for both output supervision and centroid precomputation. Settings where teacher access is partial—such as logits-only API access—preclude the extraction of intermediate hidden states required by NNM. Extensions to ensemble distillation, self-distillation, or black-box teacher settings remain directions for future work.

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A Implementation Details

A.1 Efficient Nuclear Norm Computation via Newton–Schulz

Exact computation of the nuclear norm requires a full Singular Value Decomposition (SVD). In standard deep learning frameworks, SVD is highly inefficient on GPUs as it relies on sequential routines that severely under-utilize Tensor Cores. To ensure NuNo-KD is practically scalable, we completely bypass explicit SVD. Instead, we compute the nuclear norm via polar decomposition using a higher-order Newton–Schulz iteration (Higham, 1986).

Given a projected joint matrix $\bar{M}$ (Eq. (9)), its polar decomposition is $\bar{M} = UP$, where U is semi-orthogonal (it has orthonormal columns) and P is symmetric positive semi-definite. The nuclear norm is strictly defined as $\|\bar{M}\|_* = \mathrm{Tr}(P) = \mathrm{Tr}(U^\top \bar{M})$. To approximate the polar factor U efficiently, we initialize $X_0 = \bar{M}/(\|\bar{M}\|_F + \epsilon)$ and apply a 5th-order Newton–Schulz iteration:

$$X_{j+1} = X_j \left(aI + bA_j + cA_j^2\right) \qquad (14)$$

where $A_j = X_j^\top X_j$. We use the standard coefficients $(a, b, c) = (15/8, -5/4, 3/8)$, given by the Taylor expansion underlying this iteration, and fix the number of iterations to $N_{\text{iter}} = 5$.

This 5-step iteration maps well to GPU Tensor Cores, as it relies exclusively on highly optimized general matrix multiplications (GEMMs). Once $X_5 \approx U$ is computed, the nuclear norm of $\bar{M}$ is efficiently extracted as $\|\bar{M}\|_* \approx \mathrm{Tr}(X_5^\top \bar{M})$. This architectural choice ensures that NuNo-KD introduces negligible computational overhead while guaranteeing strict numerical stability throughout training.

A.2 Hyperparameters

Table 6 reports the full set of hyperparameters used across all distillation experiments.

A.3 Ablation and Baseline Details

Component ablations. All component ablations in Table 3 share the NuNo-KD reference run and change only the indicated component. *W/o centroid anchoring* matches $\|\bar{H}_s^{(i)}\|_*$ and $\|H_t^{(i)}\|_*$ directly within each mini-batch; *frozen random* $W_{s\to t}$ keeps the projector at its random full-rank initialization; *Frobenius regression* replaces $\mathcal{L}_{\text{NNM}}$ with a Frobenius (mean squared error) loss between the projected student and the teacher hidden states.

Hyperparameter	Qwen2.5	Gemma-2
Batch Size	64	64
Learning Rate	5×10^{-6}	1×10^{-7}
LR Scheduler	Cosine	Cosine
Temperature of KD	1.0	1.0
Max Sequence Length	1024	1024
Epochs	2	2
Optimizer	AdamW	AdamW
Warmup Ratio	0.1	0.1
Loss weight λ	0.2	0.2
Skew coefficient α	0.1	0.1
Supervised layers n	4	4
Relative depth range	$[0.20, 0.85]$	
Number of centroids K	64	64
Projection dimension d' (of R)	256	256
Newton–Schulz iterations N_{iter}	5	5
Newton–Schulz coefficients (a, b, c)	$(15/8, -5/4, 3/8)$	
Centroid prepass batches	500	500

Table 6: Detailed hyperparameters used for our distillation experiments.

Gram, CKA, and direct spectrum. For the selected response-token hidden states $H \in \mathbb{R}^{m\times d}$ of a layer, let $X = H - \mathbf{1}\bar{h}^\top$ be the centered matrix. The Gram objective matches the centered, trace-normalized token Gram matrices $G = XX^\top/\|X\|_F^2$ of student and teacher with a mean squared error; the CKA objective minimizes $1 - \langle G_s, G_t\rangle_F/(\|G_s\|_F\|G_t\|_F)$, i.e., linear CKA; the direct-spectrum objective matches the sorted eigenvalues of $XX^\top/d$ with a Smooth-ℓ_1 loss. Token Gram matrices are comparable across hidden widths, so these runs use no projector, centroids, or random projection. They use the same layer pairs as NuNo-KD, average the loss over the pairs, and sample up to 64 response tokens per step, shared by teacher and student. The auxiliary weights are 10^5 (Gram) and 10 (CKA and direct spectrum); each is switched on after 100 optimizer steps and ramped up linearly over the next 200. The runs fully fine-tune the student for two epochs (learning rate 5×10^{-6}) on stored student-generated responses for the TSD-KD release of UltraInteract (79,751 examples) rather than on freshly sampled rollouts. Table 3 therefore compares objectives in their respective configurations, not as a single-factor ablation.

B Additional Analyses

NuNo-KD reshapes the spectrum, not just its scale. The nuclear norm grows under uniform rescaling, and so does the size-normalized $\|H\|_*/\sqrt{Nd}$ of Figure 3, on which NuNo-KD lies below the baselines. We therefore also measure the Frobenius-normalized nuclear norm $q(H) = \|H\|_*/\|H\|_F$, which satisfies $q(aH) = q(H)$ for

any $a \neq 0$. On raw student hidden states, NuNo-KD exceeds both Student and AMiD at all five reported layers (Figure 6), for example 4.23 versus 2.82 and 3.06 at layer 24. NuNo-KD thus spreads the spectrum more evenly while lowering the activation scale $\|H\|_F/\sqrt{Nd}$, so the increase cannot be explained by rescaling. Layers 24 and 26 lie above the last supervised layer, consistent with the propagation of the shift described in Appendix C.

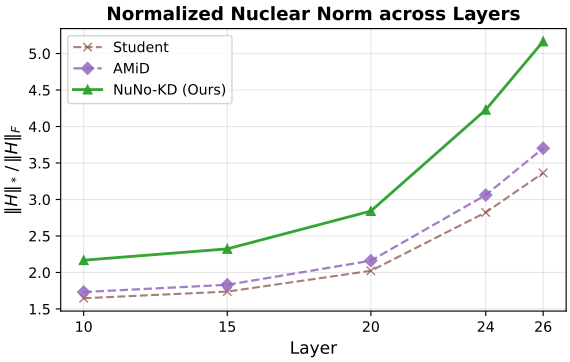


Figure 6: Frobenius-normalized nuclear norm $\|H\|_*/\|H\|_F$ of raw student hidden states (Qwen2.5-1.5B-Instruct), which is invariant to uniform rescaling, unlike the size-normalized version in Figure 3.

Matching a teacher target is stable; maximizing the norm is not. To test whether simply pushing the spectrum upward suffices, we replace the squared matching loss in Eq. (10) with an unbounded nuclear-norm maximization term and keep all other settings fixed (Table 7). Once the spectral term becomes active, around iteration 110, the maximization term diverges from -1.07×10^2 to -1.62×10^6 within 70 iterations, the KD loss rises from 1.105 to 3.219, and generations degenerate into repetitive tokens. Under NuNo-KD, the KD loss stays between 1.144 and 1.175 and the matching term stays on the order of 10^{-4}. A finite teacher-defined target thus prevents the runaway growth that maximization induces.

	KD loss		Nuclear-norm term	
Iter.	**NuNo-KD**	**Max.**	**NuNo-KD**	**Max.**
110	1.144	1.105	4×10^{-4}	-1.07×10^2
140	1.147	3.205	3×10^{-4}	-3.47×10^4
180	1.175	3.219	4×10^{-4}	-1.62×10^6

Table 7: Training dynamics of NuNo-KD and of unbounded nuclear-norm maximization (Max.).

The spectral term is cheap. On two 140 GB H200 GPUs with a per-device batch size of 8 (Table 8), NuNo-KD takes 68.6 minutes per epoch versus 66.3 for CSD and 65.1 for AMiD, i.e., 3.5% and 5.4% more time, and uses 4.2 and 6.6 GB more peak memory per GPU. The extra cost comes from the Newton–Schulz nuclear-norm estimates and the centroid-anchored matrices, and the 256-dimensional random projection keeps it small. The projector and the centroids are discarded after training, so inference cost is unchanged.

Method	**Time**	**Memory**
CSD	66.3	127.9
AMiD	65.1	125.5
NuNo-KD (Ours)	68.6	132.1

Table 8: Training cost on two H200 GPUs with a per-device batch size of 8. Time is wall-clock minutes per epoch; Memory is peak GPU memory in GB per device.

C Full Layer-Wise Matrix Entropy Distributions of the Qwen2.5-1.5B-Instruct Student

Figures 7 and 8 report the per-layer matrix entropy histograms for all 29 hidden-state layers of the Qwen2.5-1.5B-Instruct student (layer 0 is the embedding output and layers 1–28 are the outputs of its 28 transformer blocks; with $n = 4$ and the range $[0.20, 0.85]$, NNM supervises layers 5, 11, 17, and 23, at relative depths 0.18–0.82), complementing the summary view in Figure 1. Three patterns emerge across the full depth profile.

Network edges (layers 0–1 and 27–28): no separation. At the embedding output and the first block, and again at the last two blocks, all methods produce overlapping distributions with high matrix entropy (about 0.65–0.85 at layers 0–1 and 0.53–0.86 at layers 27–28). At the input end, this matches the detokenization stratum of Skean et al. (2025), where representations remain close to the token embeddings. At the output end, $\mathcal{L}_{\mathrm{SKL}}$ constrains every student in the same way through its output projection $W_o \in \mathbb{R}^{d_s \times |\mathcal{V}|}$, so the final-layer geometry converges regardless of the intermediate objective.

Compression valley (layers 2–26): consistent rightward shift. From layer 2 onward, matrix entropy collapses for every model; for the baselines it drops from about 0.8 at layer 1 to about 0.01 at

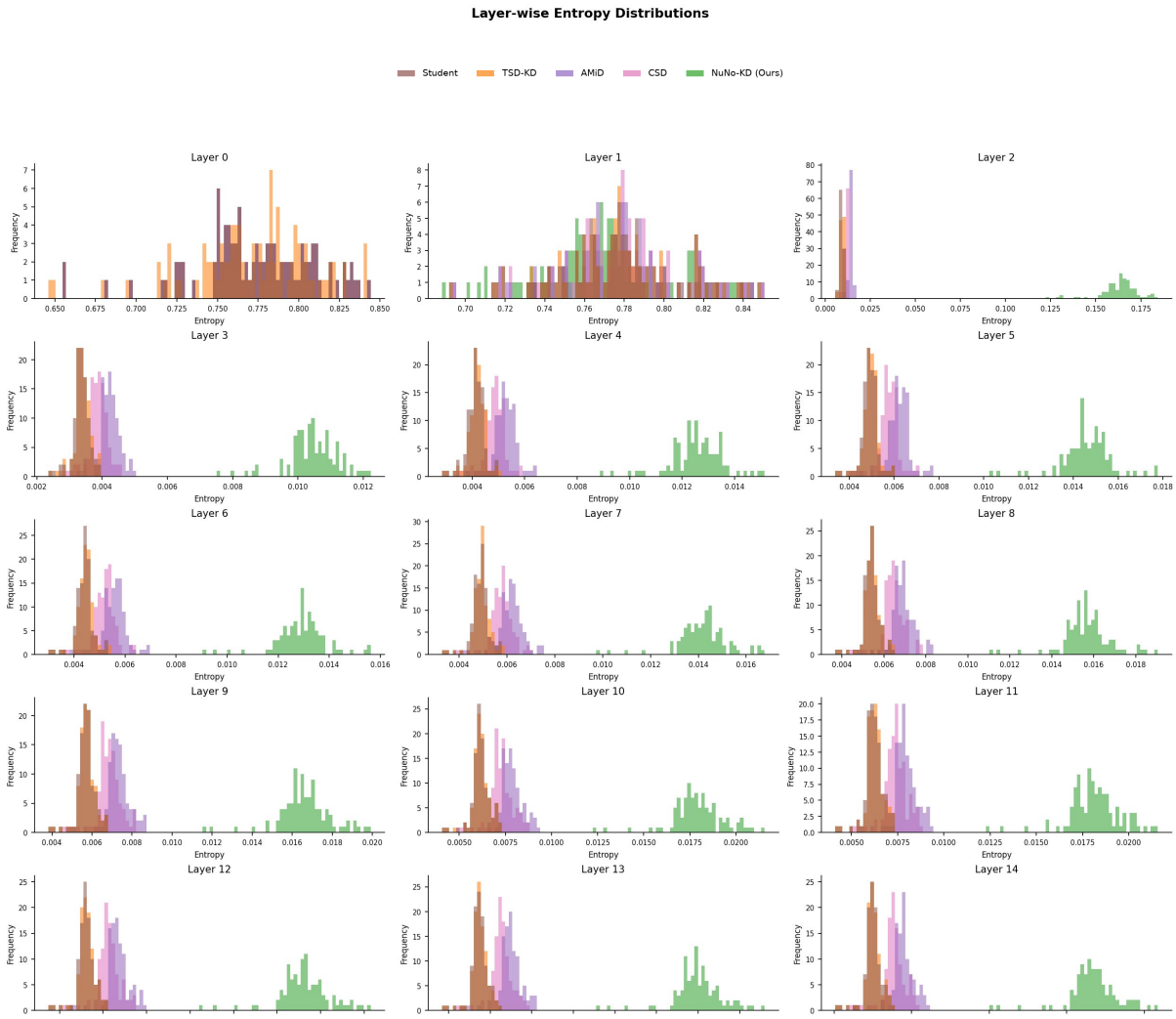


Figure 7: Layer-wise matrix entropy distributions for layers 0–14 of the Qwen2.5-1.5B-Instruct student (*continued in Figure 8*). From layer 2 onward, NuNo-KD (green) is shifted to the right of all baselines; TSD-KD overlaps Student, and CSD and AMiD shift only marginally.

layer 2. This is the compression valley described by Queipo-de Llano et al. (2026) and Skean et al. (2025). Within the valley, NuNo-KD (green) is separated from all baselines at every layer: its matrix entropy is more than ten times that of the baselines at layer 2 (about 0.16 versus 0.007–0.015) and two to three times at layers 3–26. TSD-KD overlaps Student throughout, while CSD and AMiD shift only marginally to the right, consistent with the observation that output-level objectives barely change intermediate spectral structure (Section 3.1).

Effect beyond the supervised layers. NNM is applied only at layers 5, 11, 17, and 23, yet the shift is equally visible at the unsupervised layers between them, below the first one (layers 2–4), and above the last one (layers 24–26). The gradient of $\mathcal{L}_{\mathrm{NNM}}$ at a supervised layer updates all the blocks below it, and the residual stream carries the resulting representations into the blocks above, so NNM reshapes the whole compression valley rather than isolated layers. The shift disappears only at layers 27–28, where the output-level objective dominates.

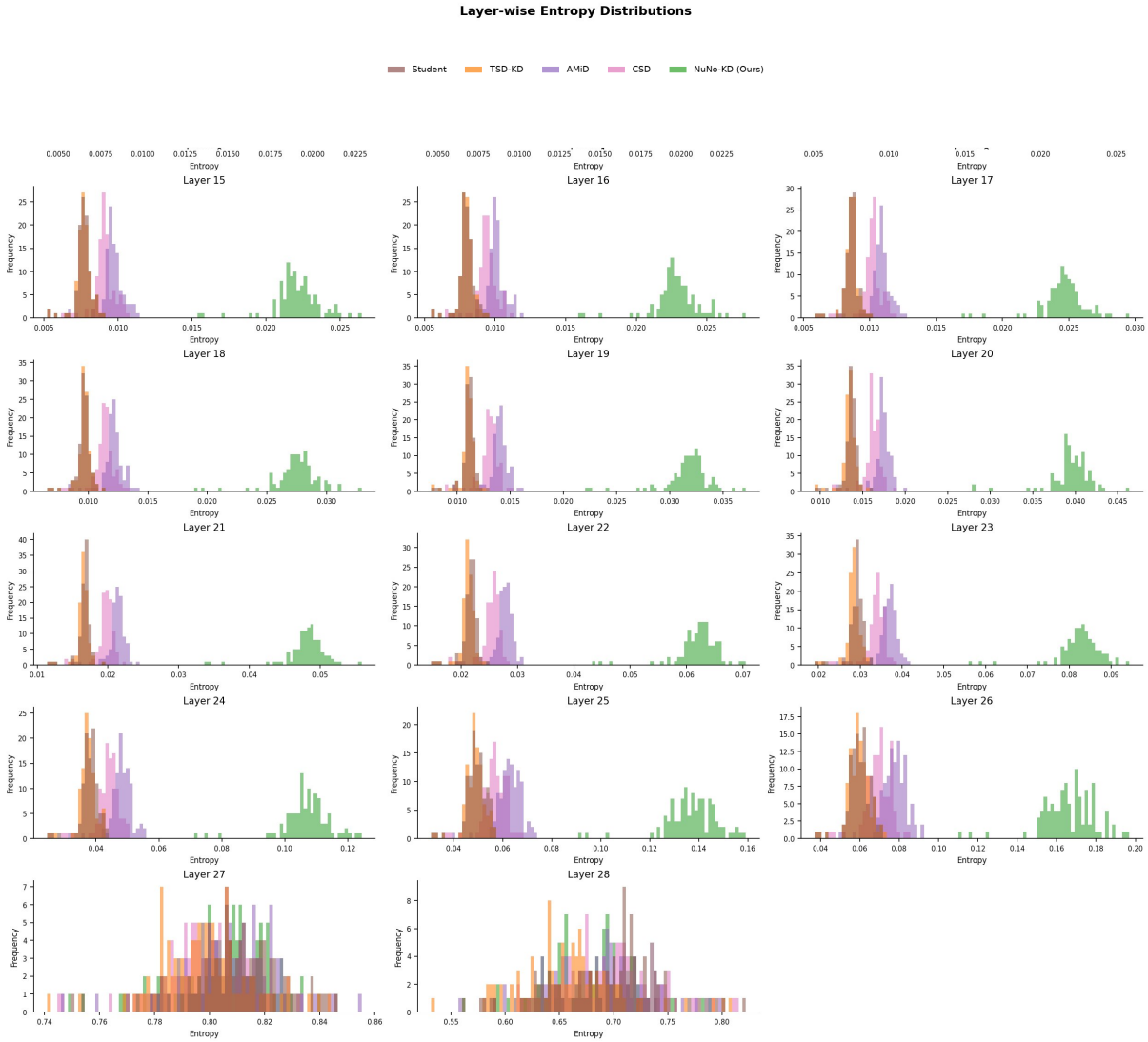


Figure 8: Layer-wise matrix entropy distributions for layers 15–28 of the Qwen2.5-1.5B-Instruct student (*continued from Figure 7*). The rightward shift of NuNo-KD persists through layer 26, beyond the last supervised layer (layer 23, relative depth 0.82), and disappears at layers 27–28, where all methods converge.